



Year 7

All about Measurement





Converting between
UNITS OF TIME

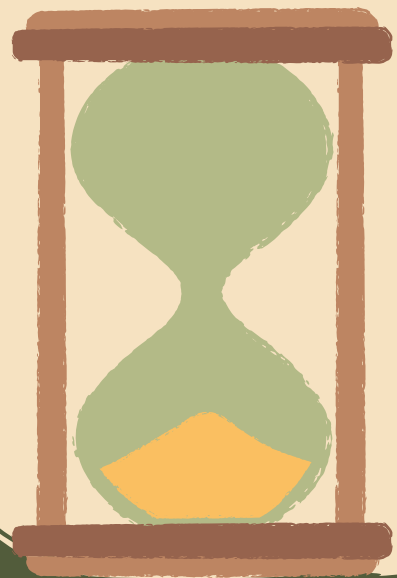


WHAT IS TIME?



Time is a tool that helps us keep track of when things happen or measure how long something takes.

We use a watch or clock to keep track of time, such as the time school starts, lunch times and bed time. We use calendars to keep track of days, months and years. It helps us stay organised and allows us to plan special events.



UNITS OF TIME

Can you match up the correct time conversions?



1 minute

1 week

1 hour

1 year

1 day

1 year

365 days

7 days

60 seconds

60 minutes

52 weeks

24 hours

UNITS OF TIME

Can you match up the correct time conversions?



1 minute

60 seconds

1 week

7 days

1 hour

60 minutes

1 year

52 weeks

1 day

24 hours

1 year

365 days

CONVERT 5 MINUTES INTO SECONDS

1 minute = 60 seconds

5 minutes = 300 seconds

60 s

60 s

60 s

60 s

60 s



CONVERT 2 MINUTES AND 30 SECONDS INTO SECONDS

1 minute = 60 seconds

2 minutes and 30 seconds = 150 seconds

60 s

60 s

30 s





CONVERT 1 HOUR AND 24 MINUTES INTO MINUTES

1 hour = 60 minutes

1 hour and 24 minutes = 84 minutes

60 minutes

24 minutes



PRACTICE QUESTIONS

Can you find the equivalent times?



1.

3 minutes and 14 seconds =

seconds

2.

2 hours and 48 minutes =

minutes

3.

5 weeks and 4 days =

days

PRACTICE QUESTIONS

Can you find the equivalent times?



1. 3 minutes and 14 seconds = 194 seconds

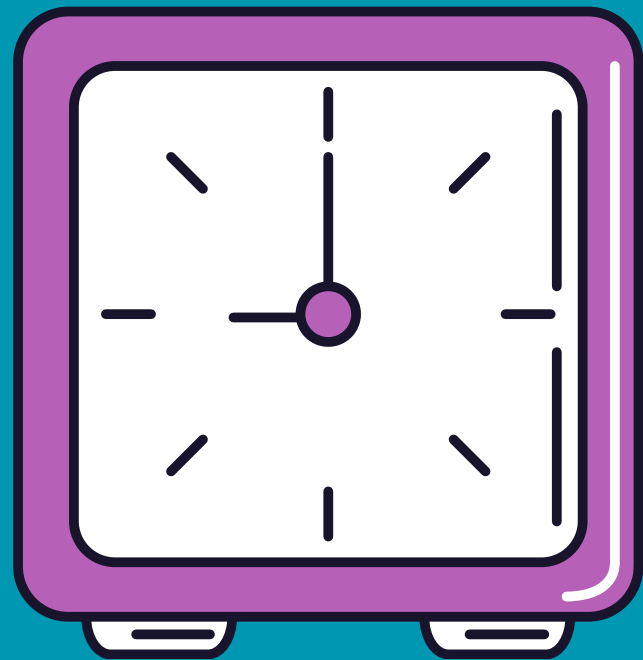
2. 2 hours and 48 minutes = 168 minutes

3. 5 weeks and 4 days = 39 days

ANSWER KEY

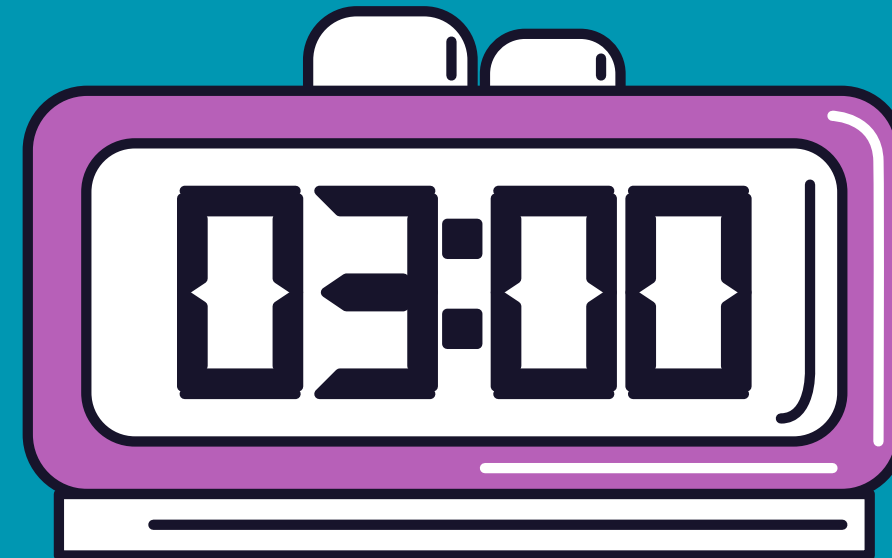
CLOCKS & TIME

ANALOGUE CLOCK



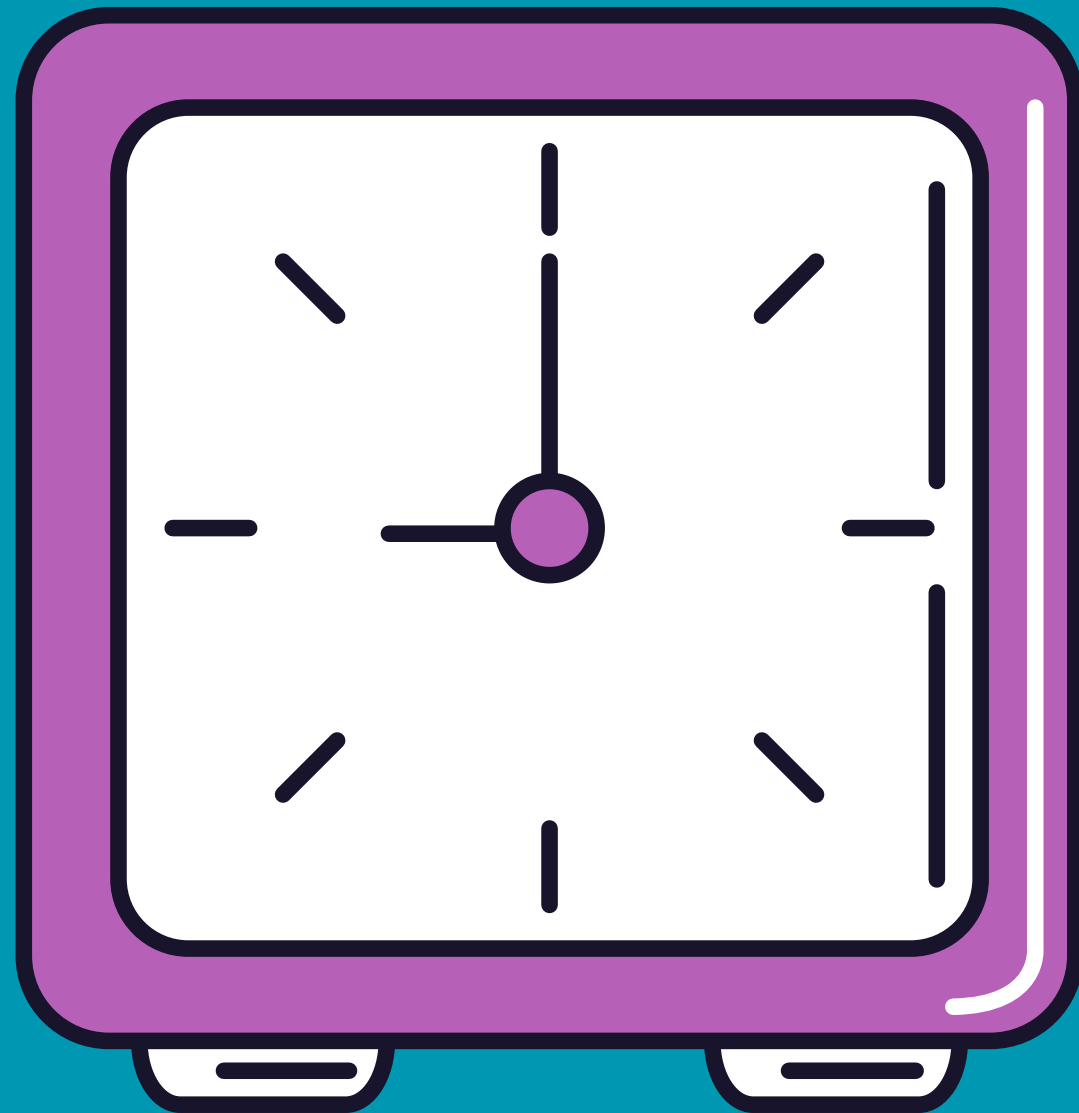
Analogue clocks
show 12-hour time.

DIGITAL CLOCK



Digital clocks can show
12- and 24-hour time.

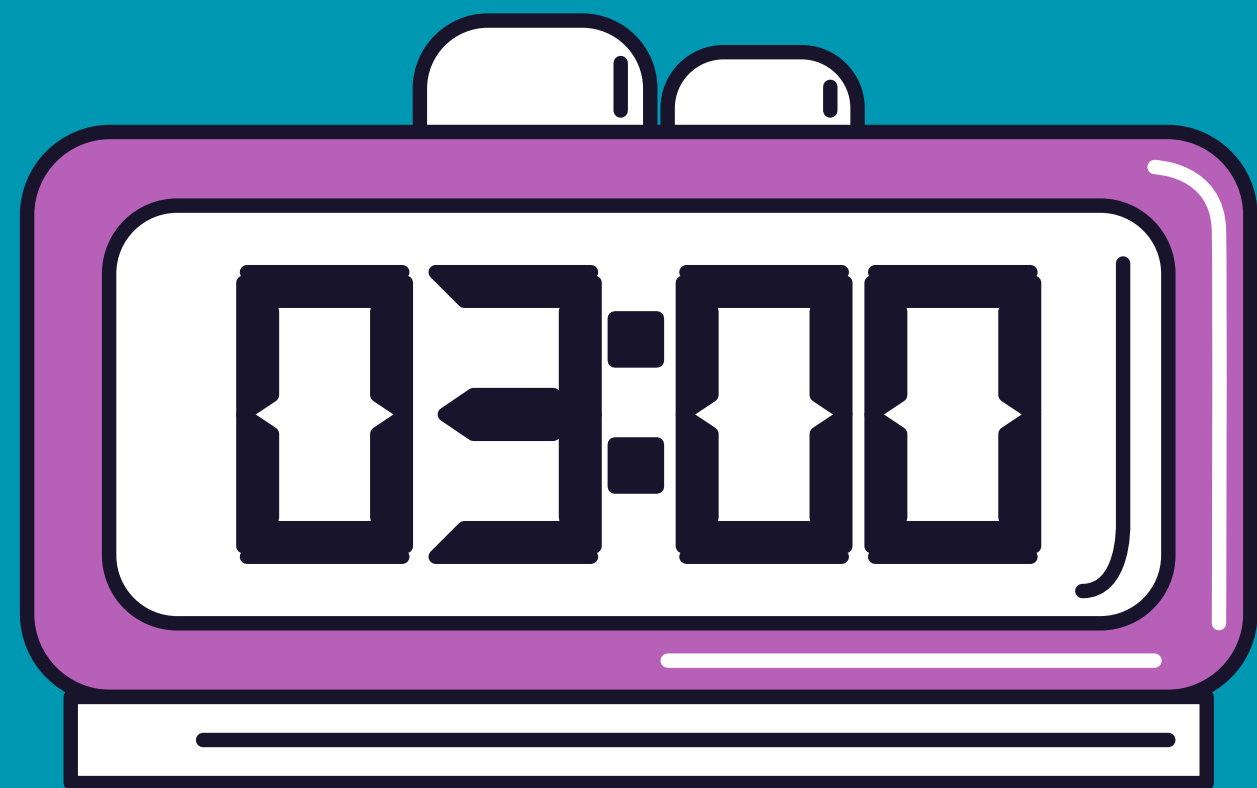
ANALOGUE CLOCK



An analogue clock shows the numbers 1 – 12. This means two full rotations represents 24 hours (one day).

When an analogue clock shows a specific time, let's say 8.30, it could mean 8.30am or 8.30pm.

DIGITAL CLOCK



An analogue clock could show time in two main ways:

12-hour time: 4.30am

24-hour time: 04.30



12-HOUR TIME

12-hour time needs to include am or pm because it breaks up the day into to 12 hour halves.

The am and pm tell us:

- It is before midday (in the morning)
- It is after midday (in the afternoon/evening)



24-HOUR TIME

24-hour time covers all 24 hours in one day so it does not need to show am or pm.

For 24-hour time:

- 0:00 – 11:59 is midnight to (almost) midday
- 12:00 – 23.59 is midday to (almost) midnight

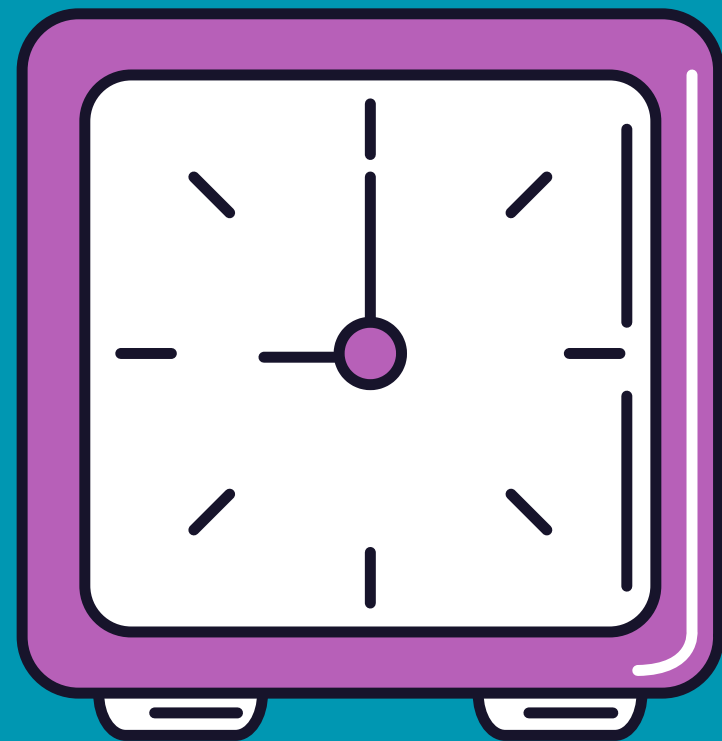


To convert 12-hour time to 24-hour time, add 12 to the hours after midday.

For example: 2:00pm is after midday.

$$2 + 12 = 14$$

This means, 2:00pm is 14:00 in 24-hour time.



To convert 24-hour time to 12-hour time, subtract 12 from the hours after midday.

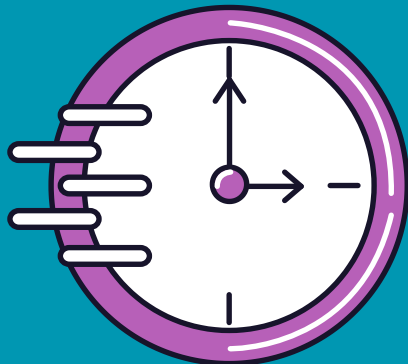
For example: 16:45 is after midday.

$$16 - 12 = 4$$

This means, 16:45 is 4:45pm in 24-hour time.

12-HOUR TIME	24-HOUR TIME
12:00am (midnight)	00:00
1:00am	01:00
2:00am	02:00
3:00am	03:00
4:00am	04:00
5:00am	05:00
6:00am	06:00
7:00am	07:00
8:00am	08:00
9:00am	09:00
10:00am	10:00
11:00am	11:00

12-HOUR TIME	24-HOUR TIME
12:00pm (midday)	12:00
1:00pm	13:00
2:00pm	14:00
3:00pm	15:00
4:00pm	16:00
5:00pm	17:00
6:00pm	18:00
7:00pm	19:00
8:00pm	20:00
9:00pm	21:00
10:00pm	22:00
11:00pm	23:00



LET'S TRY A PROBLEM

Esther was given a new digital clock for her birthday.
Her Grandma set it up to show 24-hour time.



The clock is showing 17:45. Esther has soccer training
at 6:30pm. She isn't sure if she's late!

Can you help her?



17:45 is after 12:00. This means it is after midday
(the 2nd half of the day).

To convert 17:45 into 12-hour time, we need
to subtract 12 hours.

$$17 - 12 = 5$$

This means it is 5:45pm in the afternoon.

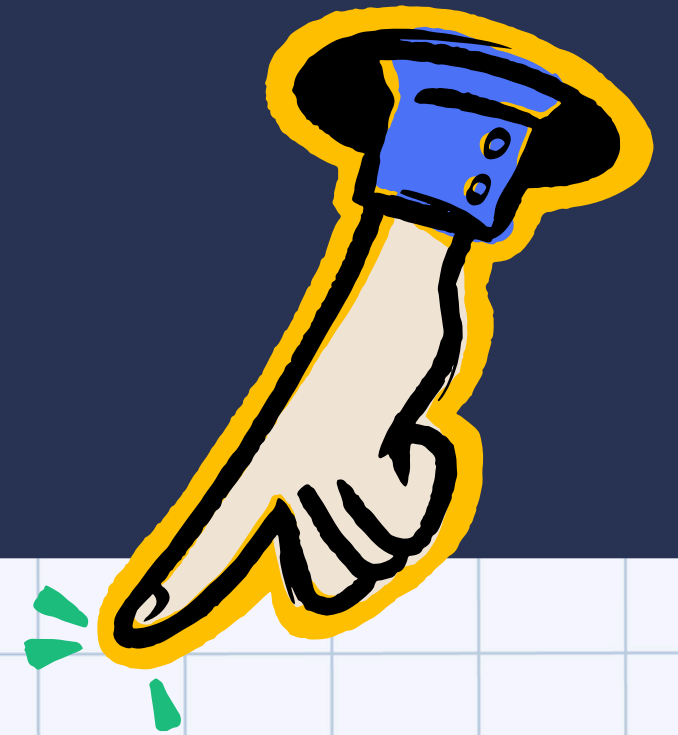
Esther isn't late for soccer training!



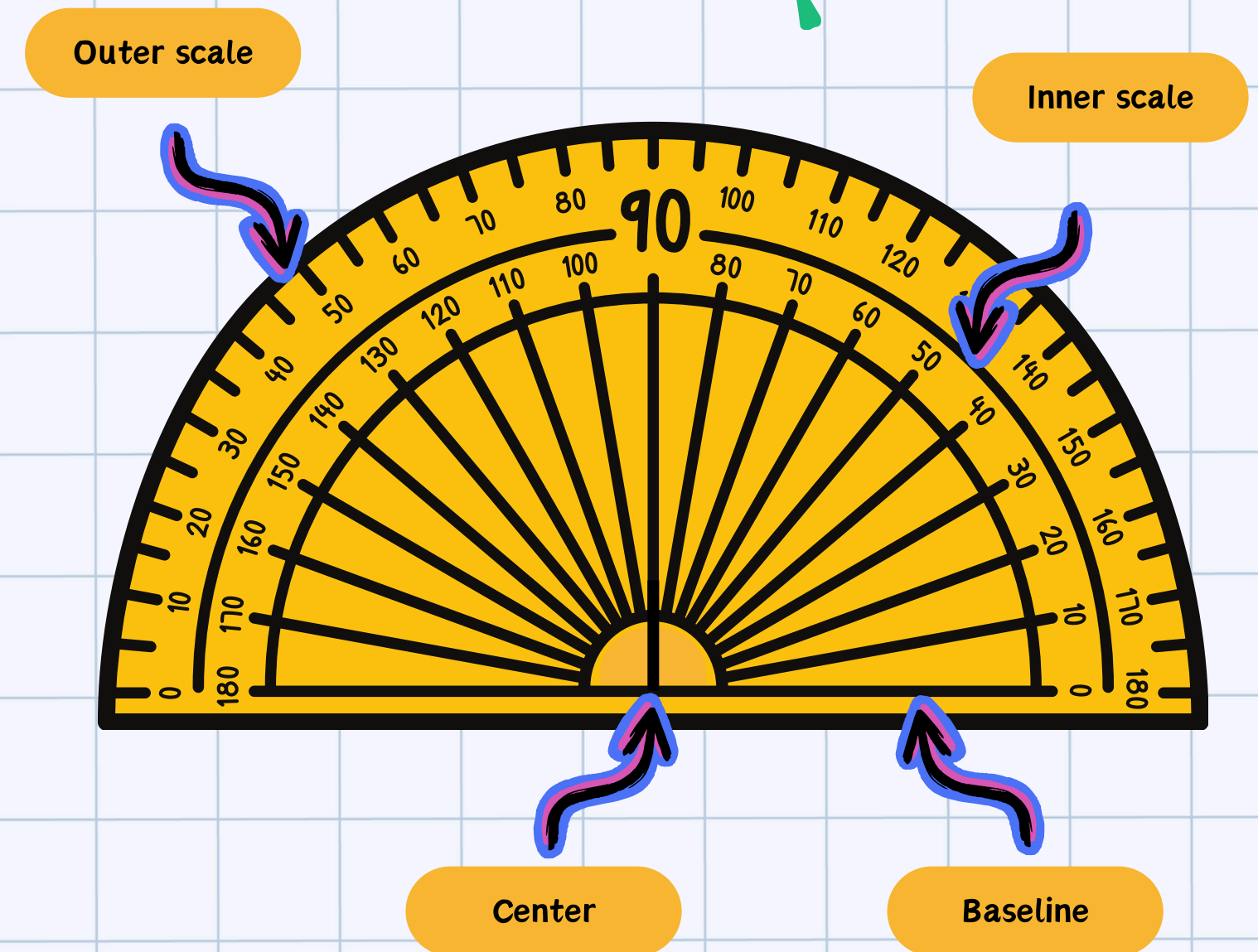
Measuring Tools



What is a protractor?

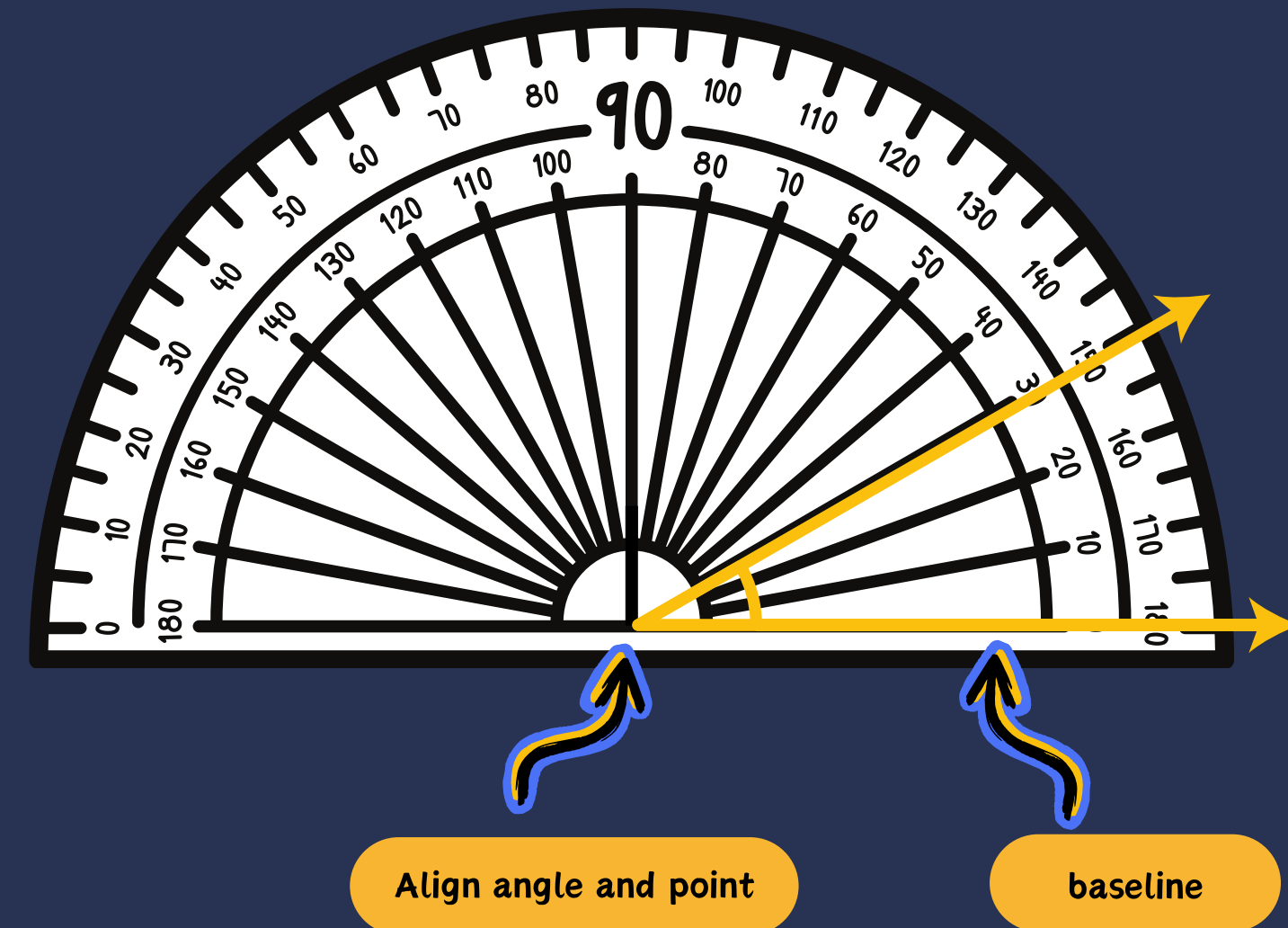


A protractor is a semi-circular or circular tool marked with degrees from 0° to 180° or 0° to 360° .



How to use a protractor?

- 1 Place the Protractor: Align the baseline of the protractor with one side of the angle, ensuring the midpoint (center hole) is on the vertex of the angle.
- 2 Align the Baseline: Make sure one side of the angle follows the baseline (0° mark) of the protractor.
- 3 Read the Measurement: Look at where the other side of the angle intersects the numbered scale on the protractor. This is the measure of the angle in degrees.

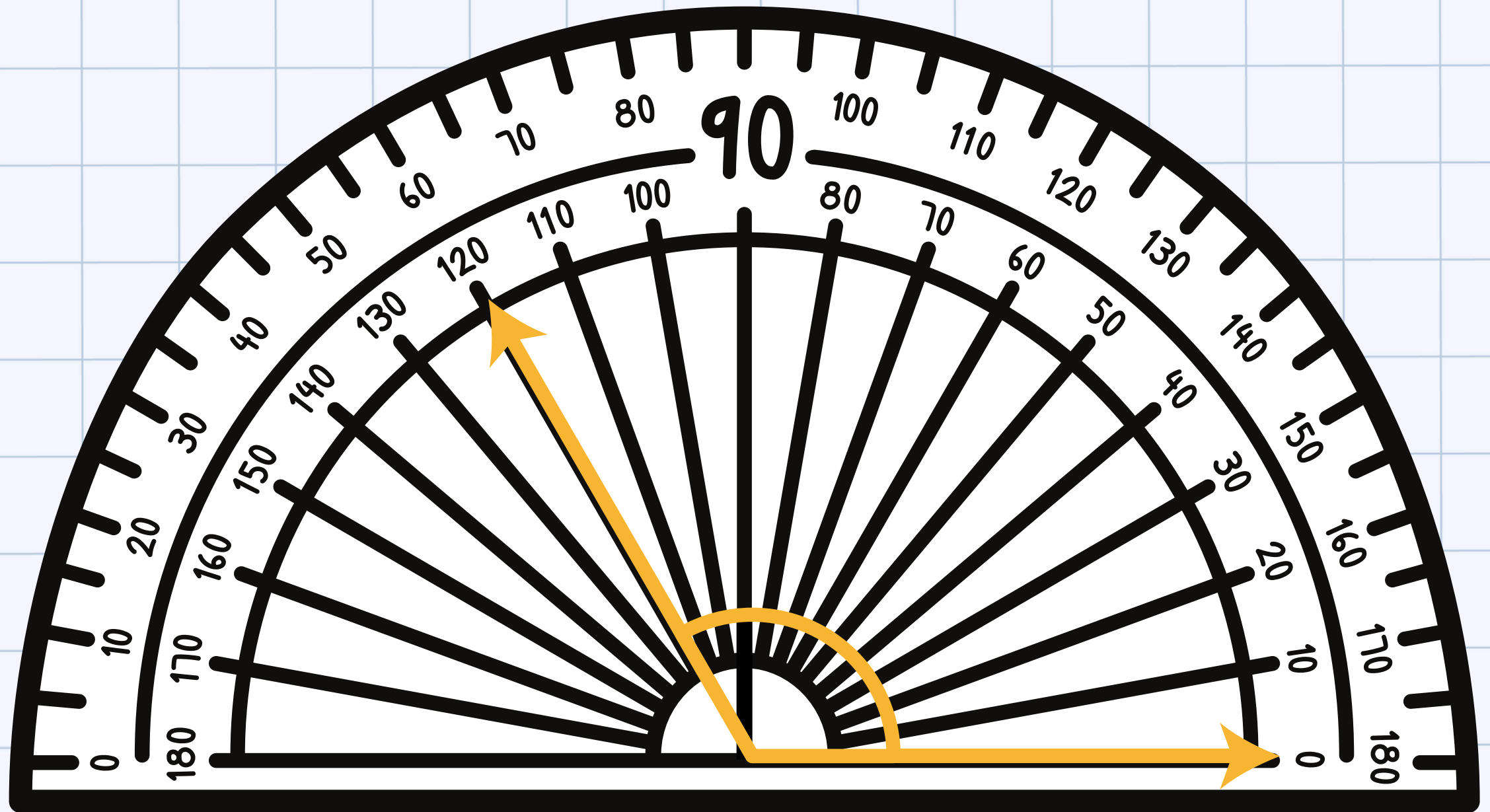


What is the measurement of this angle?



A) 120°

B) 60°

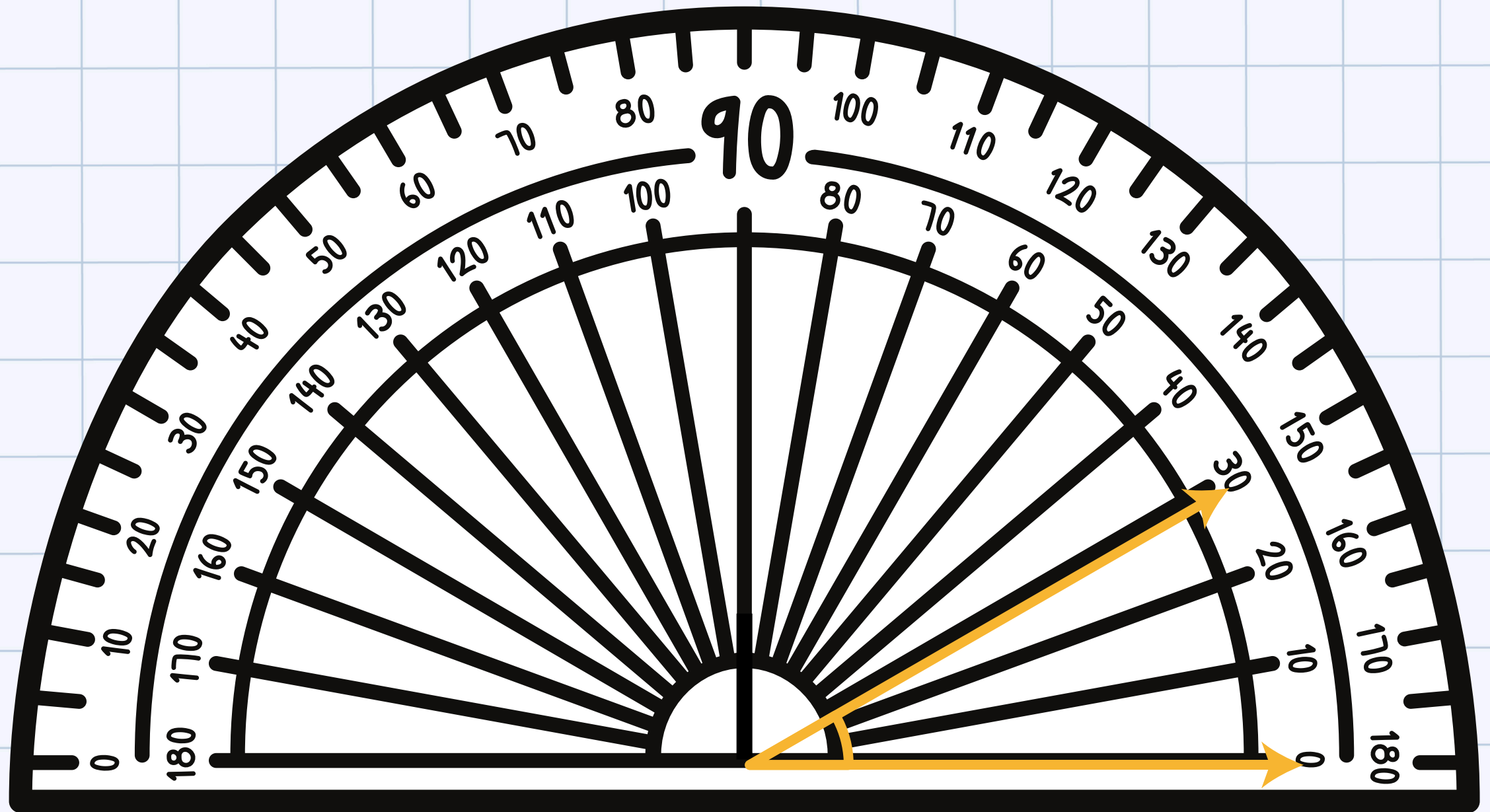


What is the measurement of this angle?



A) 150°

B) 30°

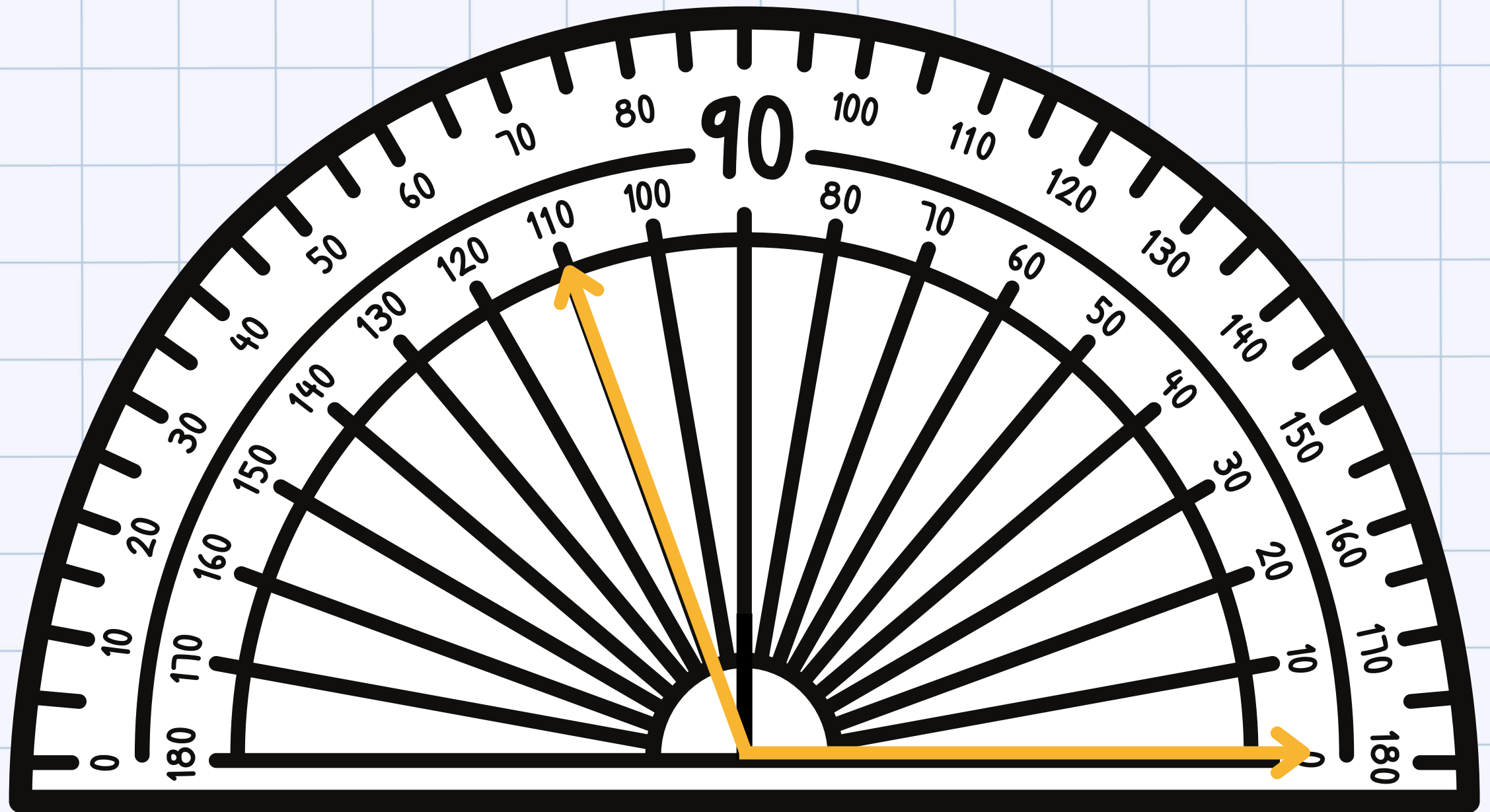


What is the measurement of this angle?



A) 110°

B) 70°

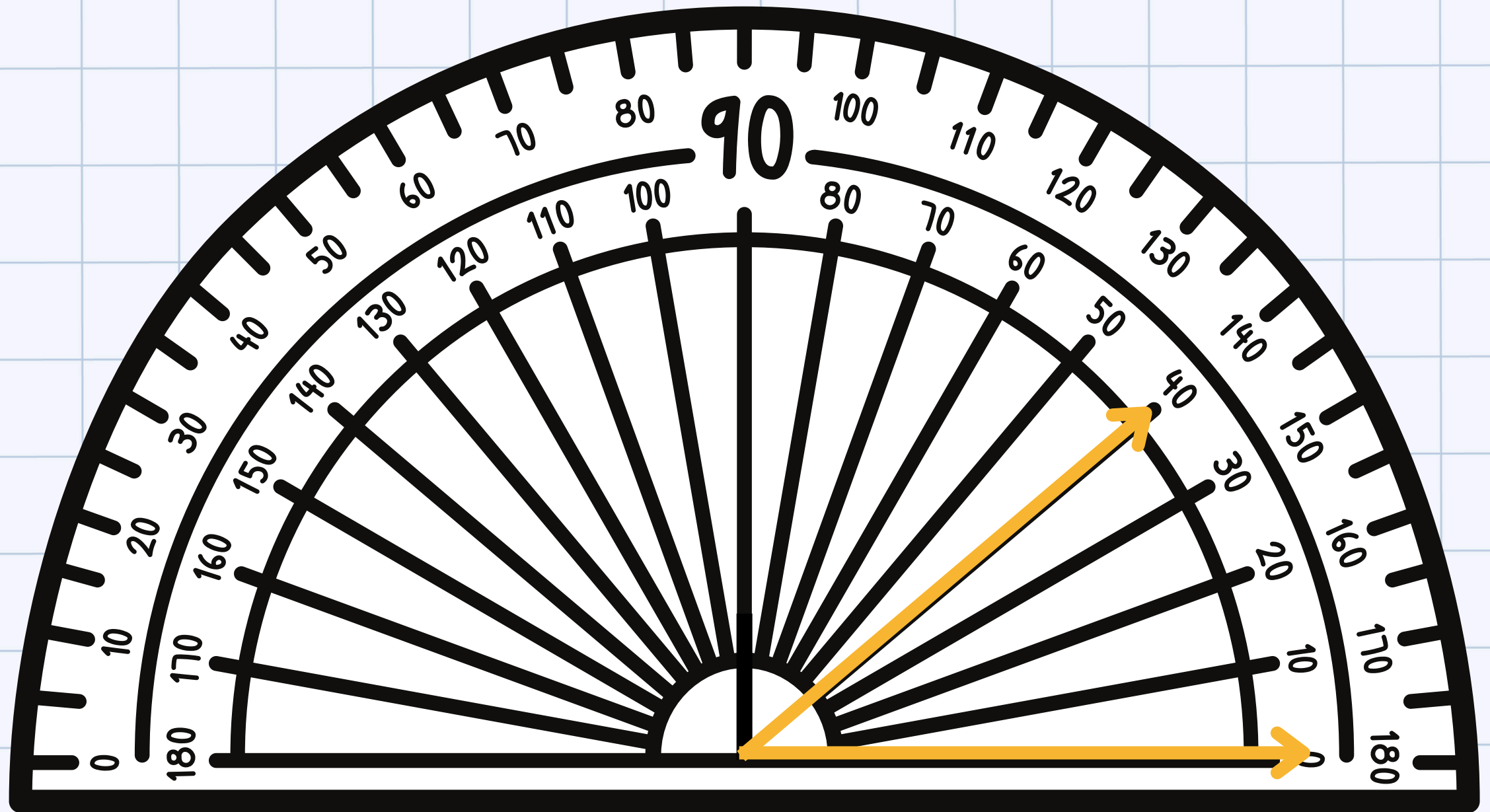


What is the measurement of this angle?



A) 140°

B) 40°

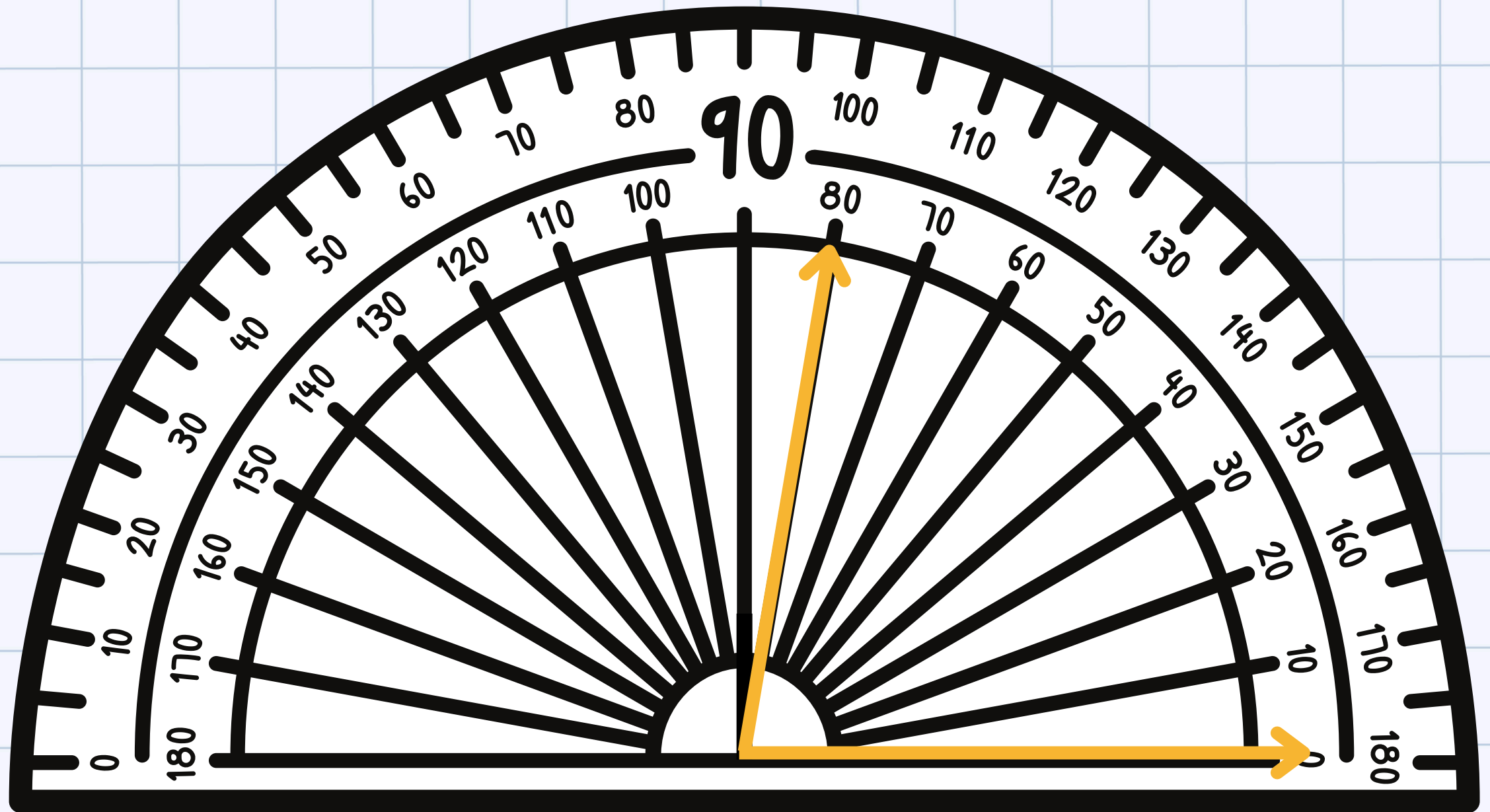


What is the measurement of this angle?



A) 100°

B) 80°

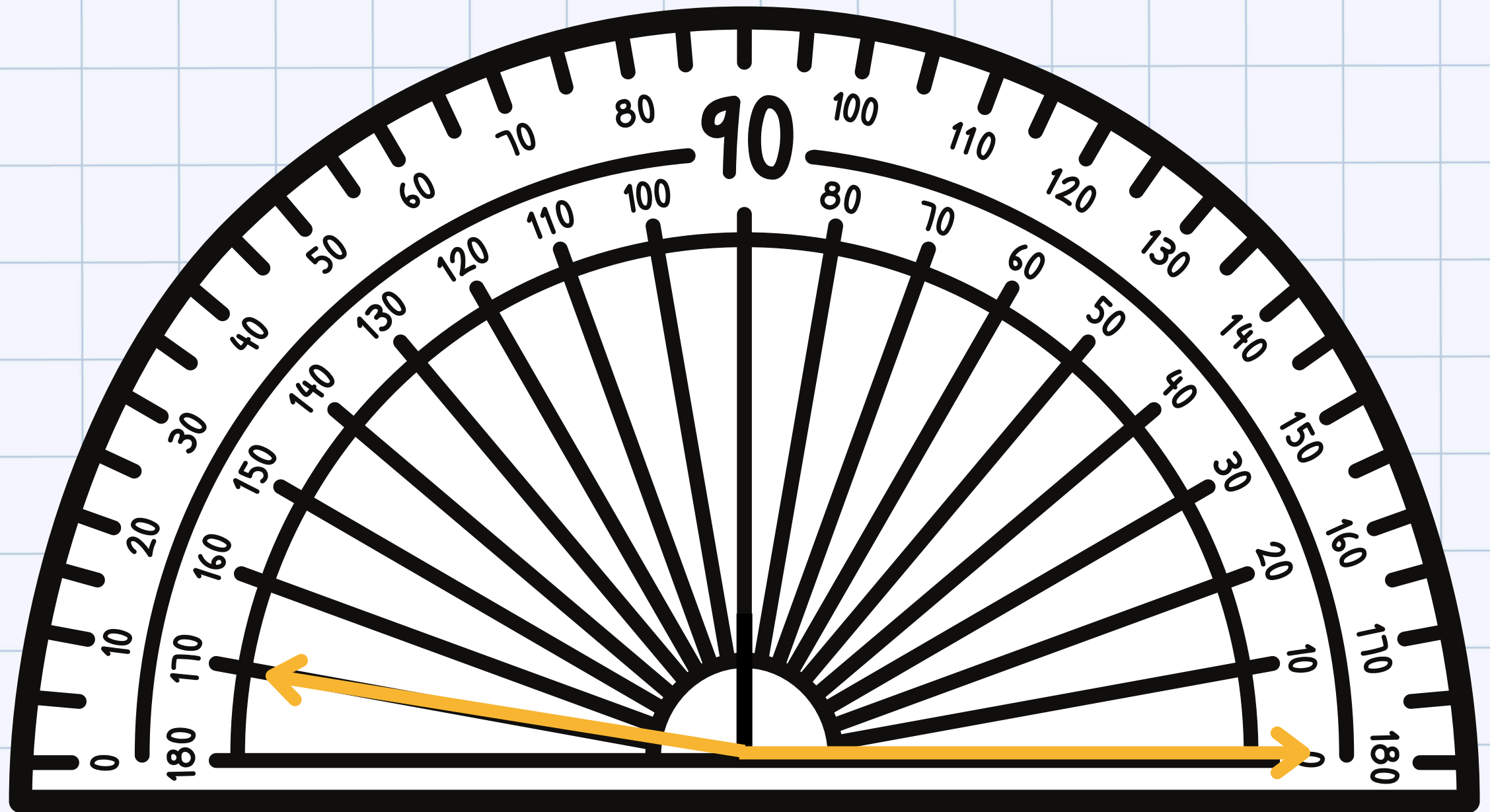


What is the measurement of this angle?

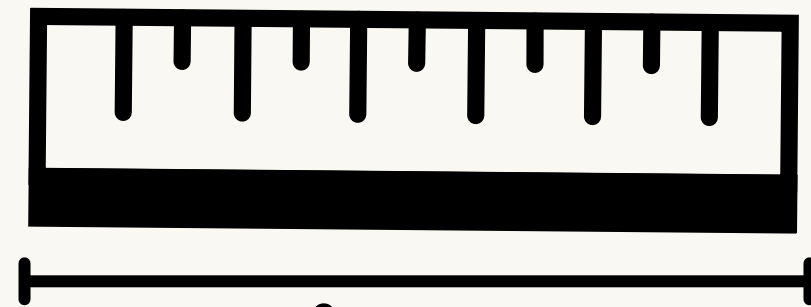


A) 170°

B) 10°



Tools:

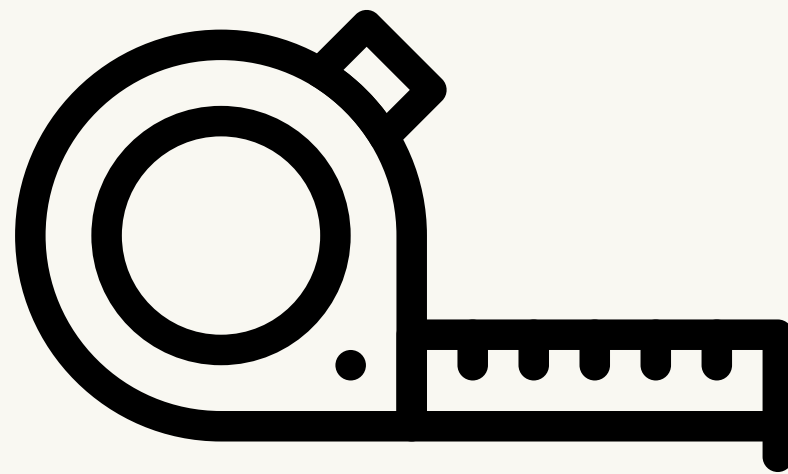


1 foot

1 foot = 12 Inches

Ruler

Measures the units
inch and foot.

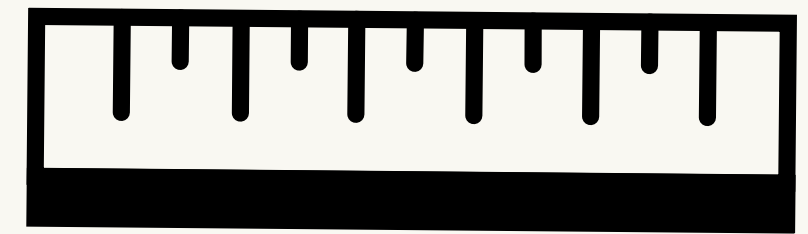


Measuring Tape

Measures the units
inch and foot.



Tools:

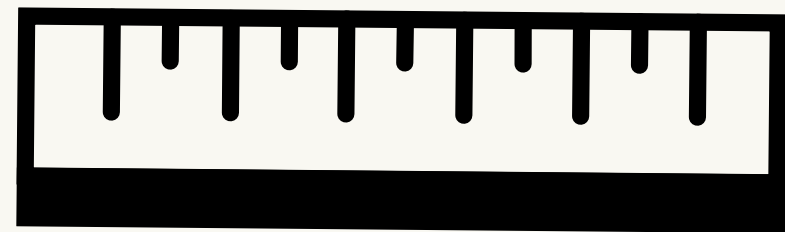


1 meter

$1\text{ m} = 100\text{ cm}$

Meter Stick

Measures the units
centimeters and meters

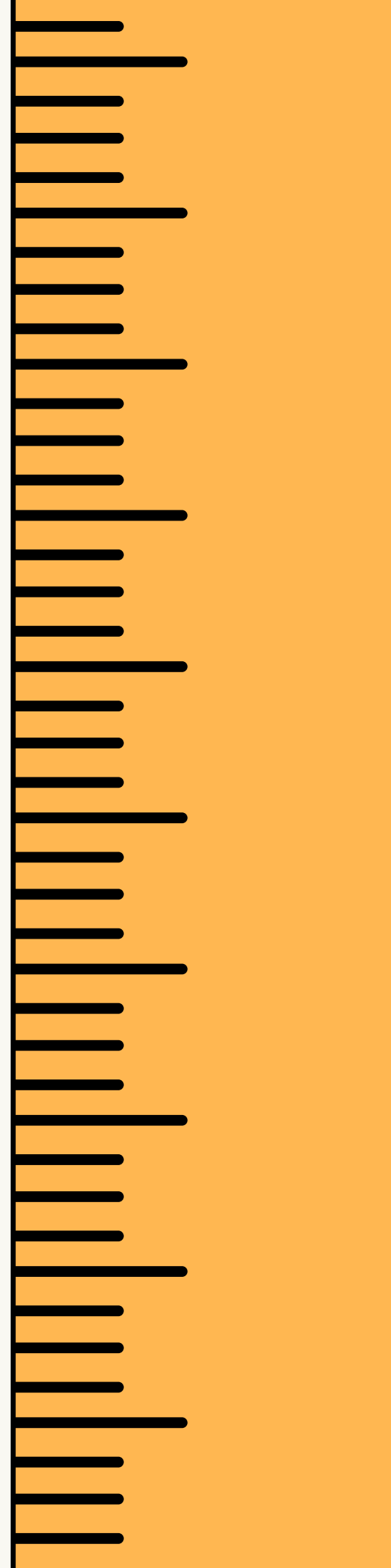


1 yard

$1\text{ yard} = 3\text{ ft}$

Yard Stick

Measures the units
yard, feet and inches



WATCH



Watches measure
duration and time.

MEASURING CUPS

Measuring cups
measure quantities
needed when
baking or cooking.



KITCHEN SCALES

Kitchen scales
measure the
weight of food.



BATHROOM SCALES

Bathroom scales
measure the
weight of an
individual.



MEASURING SPOON

A measuring spoon measures the quantity of ingredients when cooking.



THERMOMETER

A thermometer measures heat: how warm or cool something is.



**Thank you!
You did great!**

